

Vehicle Dynamics Simulation Software

User Guide

Contents

1.	Introduction	1
2.	Welcome	2
3.	Tires	3
4.	Vehicle	4
4.1.	Mass and Size.....	4
4.2.	Brakes and Differentials	5
4.3.	Aerodynamics	6
4.4.	Springs	7
4.5.	Kinematics.....	9
5.	Simulation	11
5.1.	Single run	12
5.2.	YMD	13

1. Introduction

This document serves as a user guideline to operate the Vehicle Dynamics Simulation Software. It uses a quasi-steady state numerical model based on a Bachelor's thesis written by Gustavo Drissen Silva to simulate physical phenomena related to Vehicle Dynamics.

The software directory consists of a folder (VehicleDynamics) containing an executable file (VehicleDynamics.exe) and another folder (Files) that contains all the data for the

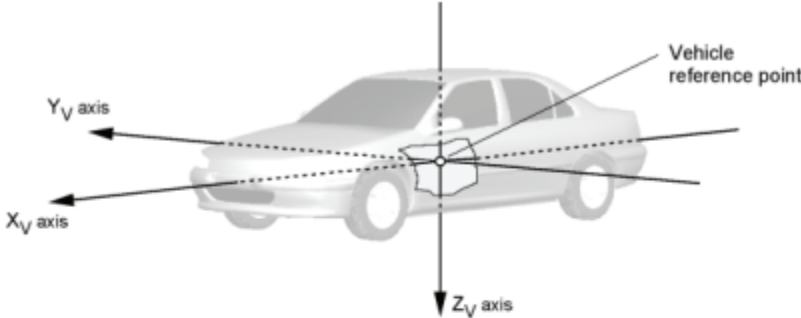
software to work. The latter shouldn't be manually altered. To open the software, simply run the executable file.

The interface of the software consists of four main tabs: Welcome, Tires, Vehicle and Simulation; and two buttons on the left: SINGLE RUN and YMD. The sections below describe each of them.

2. Welcome

The first tab in the interface gives a sign convention based on the SAE Z-DOWN axis system orientation, which is used throughout the other tabs.

Sign convention



This software uses the SAE Z-DOWN axis system orientation:

- Positive lateral forces and accelerations point to the right;
- Positive longitudinal forces and accelerations point forward;
- Positive forces point downward;
- Positive yaw moment and yaw rate rotate the vehicle clockwise from a top view perspective;
- Positive roll angle (Ψ) lowers the right side of the vehicle;
- Positive pitch angle (Φ) lowers the rear of the vehicle;
- Positive chassis slip angle (β) and steering wheel angle (δ_d) point to the right;
- Positive tire slip angles (α) point to the right;
- Positive tire slip ratios (κ) and torques happen during traction (drive).

It also contains a link to the thesis used as basis (it is recommended reading the thesis before using the software) and to this user guide.

3. Tires

The Tires tab is used to create profiles for the tires that will later be used in the simulation.

In the “Tire name” box, the user can select a tire from the list of tires in the drop-down menu, and also alter the name of the selected tire.



The image shows a user interface for managing tires. At the top, there is a label "Tire name" in blue. Below it is a dropdown menu with a white background and a thin border. The menu is open, showing the text "Hoosier 18x7,5-10 10 ps" in blue, followed by a small downward-pointing arrow. Below the dropdown menu are three buttons: "New", "Delete", and "Save". Each button is a light gray rectangle with a thin border and the text in blue. The "New" button is on the left, "Delete" is in the middle, and "Save" is on the right.

Clicking the “New” button adds a new tire to the tire list named “New tire” that will carry the parameters from the last selected tire in the drop-down menu.

Clicking “Delete” will remove the selected tire from the tire list (a message box will appear to confirm the intent of the user).

Clicking “Save” will save any changes made to the name and/or input parameters of the selected tire.

The user inserts the input parameters for the selected tire in the “Lateral”, “Longitudinal” and “Other” boxes. The “Lateral” and “Longitudinal” boxes receive the required Pacejka parameters to define the behavior of the tire. The “Other” box receives the unloaded tire radius, the rolling resistance parameters, the tire nominal load and the tire vertical stiffness, as described in the interface.

Lateral		Longitudinal		Other	
pCy1	1,00000	pCx1	1,00000	Unloaded tire radius [mm]	228,60
pDy1	-2,93292	pDx1	2,96086	Rolling resistance parameter "a" [-]	0,0258
pDy2	0,0451906	pDx2	-0,499395	Rolling resistance parameter "b" [1/N]	0,0520
pDy3	15,1193	pDx3	19,3160	Nominal tire load [N]	667,00
pEy1	-0,59049	pEx1	-0,132474	Tire vertical stiffness [N/mm]	131,36
pEy2	0,684853	pEx2	0,363558		
pEy3	0,00	pEx3	-1,98825		
pEy4	1,07811	pEx4	-0,59049		
pKy1	-58,1312	pKx1	56,5587		
pKy2	2,14319	pKx2	-0,00000285		
pKy3	3,77863	pKx3	-0,302759		
pHy1	0,001055	pHx1	0,00077354		
pHy2	-0,00033822	pHx2	0,00149605		
pHy3	0,144589	pVx1	-0,0230756		
pVy1	-0,0199972	pVx2	-0,0547334		
pVy2	0,0330034				
pVy3	0,99486				
pVy4	-2,90347				

4. Vehicle

In the Vehicle tab, the user inserts all the input parameters related to the vehicle. It is divided into five sub-tabs, which are described below.

4.1. Mass and Size

In the Mass and Size sub-tab, the user inserts the total mass of the vehicle, the lateral and longitudinal mass distribution, the height of the center of gravity, the front and rear tracks, the wheelbase, and the unsprung masses on each of the four wheels.

Total mass		Unsprung masses	
Vehicle mass (with driver) [kg]	300,00	Front left unsprung masses [kg]	11,00
Longitudinal mass distribution [%front]	48,00	Front right unsprung masses [kg]	11,00
Lateral mass distribution [%right]	50,00	Rear left unsprung masses [kg]	11,00
Height of the center of gravity [mm]	236,00	Rear right unsprung masses [kg]	11,00
		Front left unsprung mass CG height	221,69 mm
		Front right unsprung mass CG height	224,17 mm
		Rear left unsprung mass CG height	219,15 mm
		Rear right unsprung mass CG height	220,86 mm

Size		Sprung mass	
Wheelbase [mm]	1535,00	Sprung masses	256,00 kg
Front track [mm]	1250,00	Sprung masses CG height	238,50 mm
Rear track [mm]	1250,00		

The user can also visualize calculated inputs for the sprung masses, height of the center of gravity of the sprung masses and height of the center of gravity for each of the unsprung masses.

4.2. Brakes and Differentials

The Brakes and Differentials tab contains two boxes: “Brakes” and “Differentials”.

In the “Brakes” box the user can select the configuration of the brakes, being: All wheels, Front wheels or Rear wheels. The user can also select the type of mounting of the front and/or rear brakes, being: Inboard or Outboard.

Brakes	
Brake configuration	All wheels
Front brakes mounting	Outboard
Rear brakes mounting	Outboard

The “Differentials” box gives the user the same possibilities, but for the differentials. The drive configuration can be set to: All wheels, Front wheels or Rear wheels; and the mounting position of the front and/or rear differentials may be set to Inboard or Outboard. Additionally, the user can set the locking type for an existing differential, being: Open, Geared, Ramp, Trac-loc or Locked. Each locking type may require different input parameters, such as: driving and braking TBR for the Geared and Ramp types, preload for the Ramp and Trac-loc types, and differential stiffness for the Trac-loc type. The Open and Locked types require no additional parameters.

Differentials

Drive configuration

Rear wheels

Front differential mounting

Inboard

Front differential locking

Open

Front differential driving TBR

1,00

Front differential braking TBR

1,00

Front differential preload [Nm]

0,00

Front differential stiffness [Nm/Nm]

0,00

Rear differential mounting

Inboard

Rear differential locking

Ramp

Rear differential driving TBR

3,00

Rear differential braking TBR

2,00

Rear differential preload [Nm]

20,00

Rear differential stiffness [Nm/Nm]

0,00

4.3. Aerodynamics

In the Aerodynamics tab, the user can insert parameters for the vehicle downforce, downforce distribution and aerodynamic drag. All are inserted as functions of the vehicle velocity (V) in km/h.

Aerodynamics

Total downforce [N]	0,1702 * V ² +	-0,4086 * V +	0,0000
Longitudinal downforce distribution [%front]	0,0000 * V ² +	0,0200 * V +	60,0000
Lateral downforce distribution [%right]	0,0000 * V ² +	0,0000 * V +	50,0000
Total drag [N]	0,0797 * V ² +	0,0100 * V +	0,0000

4.4. Springs

In the Springs tab, the user can insert parameters for the stiffnesses of suspension springs, chassis and Anti-roll bars.

In the "Springs" box the user inserts the stiffness for the front and rear suspension springs. In the "Chassis" box the user can insert the chassis torsional roll stiffness and its longitudinal distribution, as well as a chassis torsional pitch stiffness and its lateral distribution.

Springs

Front springs
stiffness [N/mm] 42,71

Rear springs
stiffness [N/mm] 42,71

Chassis

Chassis roll
stiffness [Nm/°] 1600,00

Chassis roll
stiffness
distribution [%front] 50,00

Chassis pitch
stiffness [Nm/°] 100000,0

Chassis pitch
stiffness
distribution [%right] 50,00

In the “Create ARB” box the user can create a profile for each ARB, similarly to the tires.

Create ARB

ARB name

Front ARB 5 setups

New

Delete

Save

Add setup or remove setup

Add

50,00 [N/mm]

Remove

52,56 [N/mm]

Under “ARB name”, the user can select an ARB from the list of ARBs in the drop-down menu, and also alter the name of the selected ARB.

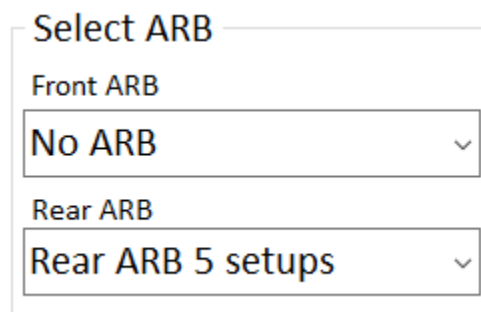
Clicking the “New” button adds a new ARB to the ARB list named “New ARB” that will carry the parameters from the last selected ARB in the drop-down menu.

Clicking “Delete” will remove the selected ARB from the ARB list (a message box will appear to confirm the intent of the user).

Clicking “Save” will save any changes made to the name and/or input parameters of the selected ARB.

There is a second drop-down menu which contains the stiffness for all the setups of the selected ARB. Clicking “Add” will add a new setup for the selected ARB with the stiffness value inserted in the numerical input box to its right. Clicking “Remove” will remove the selected setup stiffness from the setup drop-down menu of the selected ARB. An ARB can have a minimum of one setup.

In the “Select ARB” box the user can select an ARB for the front and/or rear axles based on the ARB list. There is also a standard option called “No ARB” which has no setups and adds no stiffness to the suspension.



The image shows a software interface titled "Select ARB". It contains two sections. The first section is labeled "Front ARB" and has a dropdown menu with the option "No ARB" selected. The second section is labeled "Rear ARB" and has a dropdown menu with the option "Rear ARB 5 setups" selected. Both dropdown menus have a small downward arrow icon on the right side.

4.5. Kinematics

In the Kinematics tab the user can insert the parameters that describe the dynamic behavior of the suspension.

Roll kinematics

Front roll center height [mm]	-3,3425 * ψ^2 + 0,0000 * ψ + 30,0000
Front roll center lateral offset [mm]	0,0000 * ψ^2 + 70,6650 * ψ + 0,0000
Front roll instant center lateral position [mm]	41,5000 * ψ^2 + -367,0000 * ψ + 3076,0000
Rear roll center height [mm]	0,0000 * ψ^2 + 0,0000 * ψ + 100,0000
Rear roll center lateral offset [mm]	0,0000 * ψ^2 + 22,1740 * ψ + 0,0000
Rear roll instant center lateral position [mm]	-0,0104 * ψ^2 + -27,9000 * ψ + 923,0000

Pitch kinematics

Left pitch center height [mm]	0,0000 * ϕ^2 + 0,0000 * ϕ + 80,0000
Left pitch center longitudinal offset [mm]	0,0000 * ϕ^2 + 0,0000 * ϕ + 655,9800
Right pitch center height [mm]	0,0000 * ϕ^2 + 0,0000 * ϕ + 80,0000
Right pitch center longitudinal offset [mm]	0,0000 * ϕ^2 + 0,0000 * ϕ + 879,0200
Front pitch instant center longitudinal position [mm]	0,0000 * ϕ^2 + 0,0000 * ϕ + 1000,0000
Rear pitch instant center longitudinal position [mm]	0,0000 * ϕ^2 + 0,0000 * ϕ + 1000,0000

Motion ratios

Front springs [mm/mm]	0,0000 * w^2 + 0,0024 * w + 1,4069
Rear springs [mm/mm]	0,0000 * w^2 + 0,0037 * w + 1,2458
Front ARB [mm/mm]	0,0000 * w^2 + 0,0057 * w + 1,3659
Rear ARB [mm/mm]	0,0000 * w^2 + 0,0024 * w + 1,6250

Steering

Ackermann [°]	0,0000 * δd^2 + 0,1730 * δd
Front bump steer [°]	0,0000 * w^2 + 0,0000 * w
Rear bump steer [°]	0,0000 * w^2 + 0,0000 * w

Camber

Front steering [°]	0,0015 * δ^2 + 0,0856 * δ
Rear steering [°]	0,0000 * δ^2 + 0,0000 * δ
Front bump [°]	0,0000 * w^2 + -0,0232 * w
Rear bump [°]	0,0000 * w^2 + -0,0623 * w

In the “Roll kinematics” box the user inserts parameters for the position of the front and rear roll centers and roll instant centers, all function of the roll angle (ψ) in degrees.

Similarly, in the “Pitch kinematics” box the user inserts parameters for the position of the left and right pitch centers and for the front and rear pitch instant centers, all function of the pitch angle (Φ) in degrees.

In the “Motion ratio” box, the user inserts parameters that describe the motion ratio for the front and rear suspension springs and ARBs, all function of the wheel vertical displacement (w) in millimeters.

In the “Steering” box the user inserts parameters to define the Ackermann geometry, which gives the steering angles for the front left and front right wheels, function of the driver steering wheel angle (δd) in degrees. Additionally, the user can insert parameters that describe bump steer behavior for the front and rear axles, both function of the wheel vertical displacement (w) in millimeters.

In the “Camber” box, the user inserts parameters that describe the dynamic camber for the front and rear axles. The steering camber parameters are function of the steering angles of each wheel (δ) in degrees; and the bump steer parameters are function of the wheel vertical displacement (w) in millimeters.

5. Simulation

The Simulation tab consists of two main input boxes: “Driver” and “Setup”; and two output tabs: “Single run” and “YMD”.

Driver	Setup
☒ Velocity input ☐ Corner radius input	
Vehicle velocity [km/h] 70,00	Front toe [°] 0,00
Chassis slip angle (β) [°] 1,50	Rear toe [°] 0,00
Steering wheel angle (δ_d) [°] 20,00	Front camber [°] 0,00
Longitudinal acceleration [g] 0,00	Rear camber [°] 0,00
Pedals input Cruising	Brake bias [%front] 65,00
Steering input Steering	Drive bias [%front] 0,00
	Front tires Hoosier 18x7,5-10 10 p:
	Rear tires Hoosier 18x7,5-10 10 p:
	Front ARB setup No ARB
	Rear ARB setup Setup 1/5

In the “Driver” box the user can select Velocity input or Corner radius input and insert the desired value below it. The user can also insert values for the chassis slip angle (β), driver steering wheel angle (δ_d) and the desired longitudinal acceleration for the simulation. There are two additional inputs: Pedals input and Steering input. The first can be set to: Braking, Driving, Coasting or Cruising. Braking and Driving set negative and positive longitudinal accelerations, respectively, while Cruising sets it to zero. Coasting does not impose a restriction to the longitudinal acceleration of the simulation, but rather sets all four slip ratios to zero, giving the longitudinal acceleration as an output. The Steering input selection can be set to Steering or Straight, the first allowing for a desired β and δ_d inputs, as described above, and the latter sets both β and δ_d to zero.

In the “Setup” box the user may insert toe and camber angles to the front and rear wheels, set a brake and drive bias (as long as the brake and/or drive configuration is set to All wheels), select the tires for the front and rear axles based on the tire list created in the Tires tab and select the ARB setup for the front and rear ARBs based on the setups of the selected ARBs in the Spring sub-tab of the Vehicle tab.

5.1. Single run

The Single run tab gives the user the outputs for the single run simulation. By clicking both the SINGLE RUN or YMD button on the left of the interface runs a single run simulation, but the latter also runs a YMD simulation.

The top section of the tab displays a table with outputs for each tire. The bottom section displays outputs for the vehicle as a whole, as well as some simulation outputs, like the number of iterations of the main solver. Also, the Current bias should be equal to the input brake bias for braking conditions or to the input drive bias for driving conditions.

Single run	YMD			
	Front Left	Front Right	Rear Left	Rear Right
Vertical load	1002,38 N	844,61 N	921,16 N	980,23 N
Longitudinal tire force	0,00 N	0,00 N	311,55 N	312,20 N
Lateral tire force	1144,03 N	985,49 N	-890,46 N	-970,70 N
Longitudinal chassis force	-96,83 N	-82,87 N	285,97 N	284,98 N
Lateral chassis force	1140,27 N	982,27 N	-890,46 N	-970,70 N
Slip ratio	0,00%	0,00%	0,80%	0,76%
Slip angle	1,86°	1,86°	-1,41°	-1,41°
Torque	0,00 Nm	0,00 Nm	69,04 Nm	69,04 Nm
Yaw moment: 3057,41 Nm Roll angle: -0,21° Yaw rate: 2,57 °/s Pitch angle: 0,20° Longitudinal acceleration: 0,00 g Iterations: 6 Lateral acceleration: 0,09 g Current bias: 0,00% Radial acceleration: 0,08 g Velocity: 70,00 km/h Longitudinal force: 0,03 N Corner radius: 469,02 m Lateral force: 261,38 N				

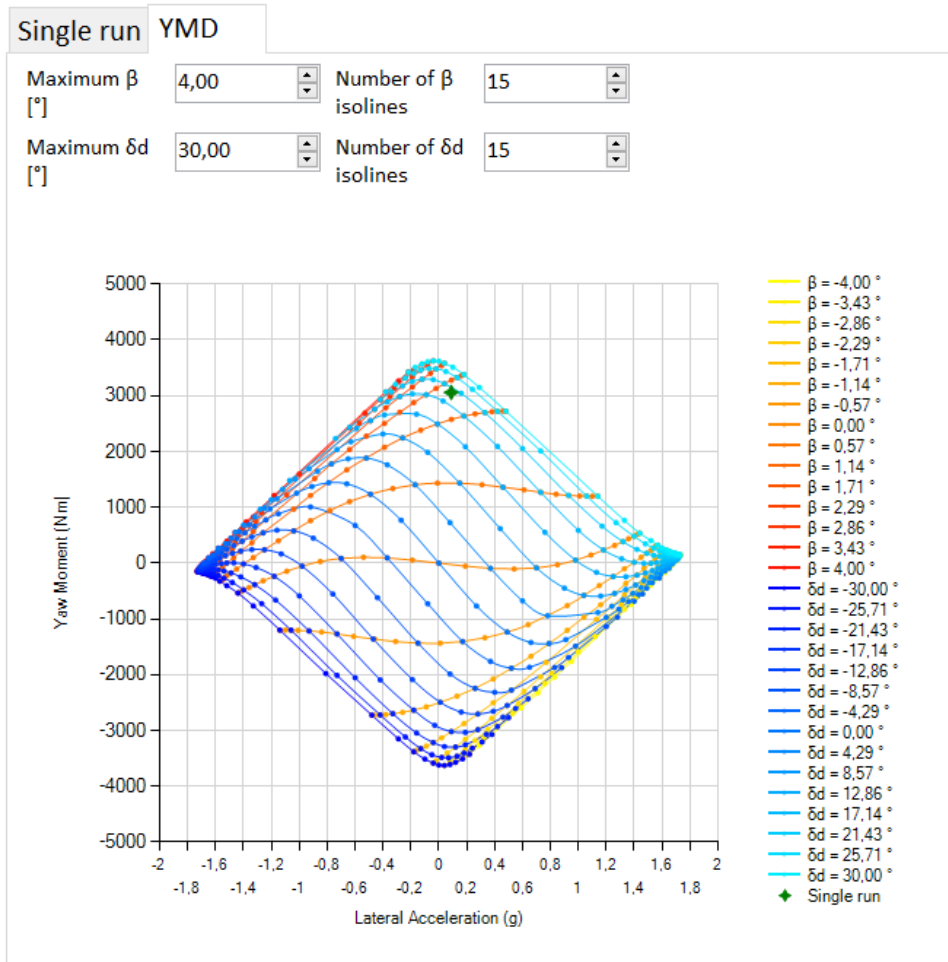
If the vehicle is not capable of achieving the desired conditions, the simulation will fail and a message box will appear to warn the user.

5.2. YMD

In the YMD tab is the YMD plotting tool.

The user can define the maximum β and δ_d for the simulations, as well as the number of β and δ_d isolines.

Clicking the YMD button on the left of the interface will run a YMD simulation and plot the YMD for the desired input parameters. It also runs a Single run simulation and plots it in the diagram. Hovering the cursor over a point on the diagram will display some outputs for that condition.



Conditions that cannot be achieved by the vehicle will cause that specific simulation to fail and the point will not be plotted in the diagram.